

Living Scroll: Multi-User Flashlight Interaction for Personalised Cultural Heritage Storytelling in Shared Space

Yutong Zhang
School of Engineering
University of Warwick
Coventry, United Kingdom
Yutong.zhang.6@warwick.ac.uk

Xinyue Yao
Department of Anthropology
University College London
London, United Kingdom
ucsaxy6@ucl.ac.uk

Tianxing Zhou
School of Communications and Arts
University of Queensland
Brisbane, Queensland, Australia
tianxing.zhou@uq.edu.au

Chengkai Xu
Department of Anthropology
University College London
London, United Kingdom
chengkai.xu.23@ucl.ac.uk



Figure 1: Living Scrolls: visitors use IR flashlights to reveal animated historical figures on a shared projected scroll while receiving personalised spatial audio.

Abstract

Co-located museum visitors make meaning socially, yet personal audio guides often isolate them. We present Living Scrolls, a low-cost (\$165) multi-user projected heritage system that lets 5–6 visitors “reveal” animated figures with IR flashlights while receiving personalised spatial audio. Each flashlight is frequency-encoded and tracked via filtered camera sensing and FFT, enabling user-specific audio and <25 ms end-to-end latency. A 5-day in-situ deployment (N=93) revealed emergent collaborative storytelling and social sharing, suggesting a scalable pattern for shared-yet-personalised heritage narration.

CCS Concepts

• **Human-centered computing** → Collaborative interaction; Auditory feedback; Empirical studies in HCI; • **Applied computing** → Media arts.

Keywords

Interactive museum exhibits, Infrared tracking, Multi-user interaction, Projection-based interaction, Shared experiences, Cultural heritage

ACM Reference Format:

Yutong Zhang, Tianxing Zhou, Xinyue Yao, and Chengkai Xu. 2026. Living Scroll: Multi-User Flashlight Interaction for Personalised Cultural Heritage Storytelling in Shared Space. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798323>



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2281-3/26/04
<https://doi.org/10.1145/3772363.3798323>

1 Introduction

Cultural heritage meaning-making is inherently personal—visitors draw different historical memories from the same painting [4]—yet museum visiting is also social: over 54% attend with family or

friends, sharing discoveries and negotiating interpretations [11, 15]. This creates a core tension: how can individuals pursue personalised narrative trajectories while sustaining shared storytelling among co-located visitors?

Existing technologies address parts of this problem but fail to support multi-threaded narratives in shared space. Audio guides personalise interpretation but suppress visitor conversation, fragmenting narration into isolated channels [9]. VR/AR creates immersive storyworlds yet disrupts the collective ritual of “standing together before the artefact” [6]. Shared-surface projection systems have shown promise for collaborative heritage exploration [7], and flashlight-metaphor devices in particular have demonstrated low learning curves and social legibility in museum contexts [13, 16]. However, prior flashlight systems either support only single-user interaction [13] or rely on device-side marker recognition that conflates collaboration with co-viewing around a single device [16], leaving each visitor without a persistent identity on the shared canvas. No existing system enables multiple co-located visitors to simultaneously pursue distinct narrative trajectories—each receiving personalised audio—while sustaining shared visual co-presence.

We introduce *Living Scrolls*, enabling 5–6 co-located visitors to explore a projected classical painting together, each triggering personalised spatial-audio narratives while maintaining shared visual co-presence. The projection initially shows landscape without human figures; visitors hold custom infrared Flashlights that render real-time spotlights (850 nm emission invisible to human vision). As visitors sweep the spotlight, animated figures gradually appear—vendors, monks, scholars—as historically grounded re-narration. When the spotlight dwells on a narrative ROI, corresponding audio plays through that visitor’s headphones, while others see the spotlights and revealed animations, sustaining shared exploration.

Technically, each flashlight modulates near-infrared emission at a unique frequency (10, 15, 20 Hz, etc.), adapting TUIC’s time-domain encoding [17]. A 70 fps camera with an 850 nm filter identifies who illuminates where via FFT analysis; Unity renders spotlights, triggers animations, and routes audio via WebSocket (<25 ms latency). Built from low-cost components (≈USD 165 vs. OptiTrack’s >USD 10,000), the system is designed for replication by resource-constrained institutions. We deployed it in a Buddhist temple-heritage setting using *The Kangxi Emperor’s Southern Inspection Tour, Volume 11*, evaluating through five-day in-the-wild deployment (20 professional reviewers; 73 participants). We observed emergent social patterns (guide–learner dynamics, collaborative exploration, social-media sharing), indicating potential beyond technical demonstration.

This work contributes: (1) design considerations derived from deployment, including routing personalised audio through a shared visual layer to preserve co-presence, and leveraging embodied metaphor as a cross-cultural interaction scaffold; (2) a low-cost multi-user prototype (5–6 users; 92% recognition accuracy; <25 ms latency); (3) field insights from a five-day in-the-wild deployment revealing emergent social dynamics; and (4) an open-source package (hardware schematics, software pipeline, calibration utilities) to support replication in resource-constrained institutions.

2 System Design

2.1 Design Goals: From Narrative Needs to Technical Constraints

Living Scrolls enables multiple visitors to pursue personalised narrative exploration in parallel while maintaining a shared context for co-present discovery. This dual requirement, individual agency and collective presence, drives our design.

We follow three principles. First, we preserve co-presence through large-scale projection rather than head-mounted displays or mobile screens, so visitors can see one another’s actions and sustain collaborative meaning-making [11]. Second, we distribute narration spatially: distinct regions of interest (ROIs) trigger different audio narratives, supporting self-directed paths through the scene in line with constructivist museum learning [10]. Third, we use an embodied flashlight metaphor to frame visitors as active discoverers who “uncover” history, rather than passive viewers.

These principles imply concrete constraints: the system must support 5–6 concurrent users without degradation, keep end-to-end latency below 50 ms to preserve narrative flow, and remain under USD 200 to enable reproducible deployment in resource-constrained settings (Figure 2a).

2.2 Narrative Space Design: One Scroll, Four Threads

We selected *The Kangxi Emperor’s Southern Inspection Tour, Volume 11* [8], a seventeenth-century handscroll depicting a major East Asian Buddhist temple complex and surrounding urban life. Animated details follow a historically grounded re-narration approach, deriving from primary sources (local gazetteers, Ming–Qing travelogues) and art-historical analysis of costume and architectural cues: vendor gestures draw on documented marketplace practices, and character dialects reflect migration patterns in the temple district.

We segmented the painting into four ROIs, each mapped to a historical thread: the Glazed Pagoda ROI (architectural achievement and religious significance); the Marketplace ROI (commercial networks and scripture circulation); the Pilgrims’ Avenue ROI (cultural encounter among European missionaries, Japanese monks, and local travellers); and the Literati Gathering ROI (elite culture through poetry, music, and board games). Each ROI triggers a 2–3 minute spatial-audio segment with embedded narrative anchors; visitors choose their own order, supporting self-directed meaning-making [11].

2.3 Interaction Design: The Flashlight Metaphor

We adopt the flashlight as the primary exploration device, extending prior single-user heritage systems [13] to multi-user narrative activation (Figure 2b).

Interaction unfolds as follows: the projection initially shows only architecture and landscape, with human figures intentionally absent to create anticipation. When a visitor points a custom near-infrared flashlight at the surface, the system renders a simulated spotlight at the corresponding location—although the infrared emission is invisible, we display a crisp circular gradient matching visitors’ mental model of a flashlight beam. As the spotlight moves, characters fade

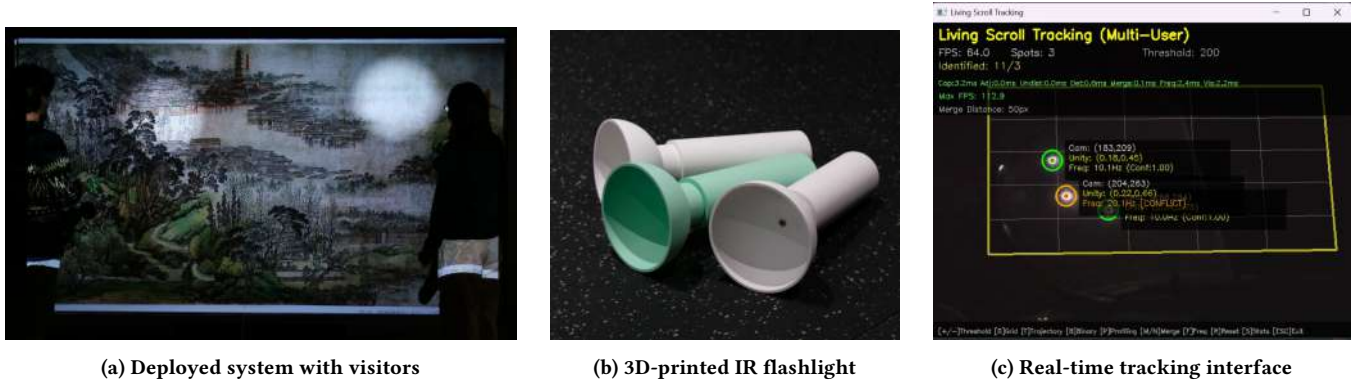


Figure 2: Living Scrolls system overview

in via smooth opacity transitions (vendors, monks, scholars), evoking a sense of “revealing” history. If the spotlight dwells within an ROI for 1.5 s, spatial-audio narration plays through that visitor’s headphones. Crucially, spotlights and revealed animations are visible to all participants, transforming individual discoveries into shared moments.

The spotlight uses a soft-edged gradient with a diameter of roughly one-eighth the projection width, balancing legibility with preservation of fine detail. We provide two physical forms: a standard flashlight and an optional red Chinese lantern with plum-orchid-bamboo-chrysanthemum motifs. Both forms are technically identical; the lantern enhances cultural immersion by inviting closer engagement consistent with the historical use of lanterns as near-held light sources, though it remains entirely optional and functions equivalently at any distance.

2.4 Feedback Design: Visual and Auditory Responses

Living Scrolls couples visual and auditory feedback to support discovery.

Visual. We implement a two-layer mask in Unity. The base layer contains the scene with original figures digitally removed and replaced by Spine 2D character animations (vendors, pilgrims, musicians). Animations are hand-drawn (Procreate), assembled into loops (After Effects), and aligned to costumes and poses in the source painting using historical references. The top layer is the figure-erased scroll as an opaque mask. Given infrared-tracked flashlight coordinates, Unity renders a circular transparent window whose position and radius update in real time, revealing the animated layer beneath via short opacity blend.

Auditory. We render spatial audio per visitor, calibrated to position and orientation. Audio comprises (i) a continuous ambient bed (market noise, bells, water) whose intensity varies with distance to each ROI, and (ii) an interactive narrated track triggered after the spotlight dwells within an ROI for a threshold duration. Assets are processed in Audition and exported as binaural stereo. Each segment includes an opening cue (discovery reward) and a closing cue with guidance to subsequent areas. As users approach an ROI centre, narration follows an inverse-square distance model and stereo panning simulates source direction.

2.5 Technical Implementation: Frequency-Encoded Tracking

Living Scrolls requires reliable user–spot association without wearables, surface instrumentation, or constrained movement. We adapt TUIC’s frequency-domain encoding to optical tracking [17]. Each flashlight uses an ESP32 to PWM-modulate an 850 nm near-infrared LED at a unique frequency (5, 10, 15, 20, 25, 30 Hz). Frequencies are spaced by ≥ 5 Hz to limit spectral leakage and kept below 35 Hz so our 70 fps stream captures multiple cycles per window. Emission (10 mW) is diffused at the flashlight head; at typical viewing distances (1–3 m) it remains below eye-safety limits.

A global-shutter industrial camera (400 fps capability) runs at 640×480 and 70 fps, viewing the surface through a narrowband 850 nm filter (± 10 nm) to suppress the projected image and ambient light. The 70 fps operational rate reflects current Python pipeline throughput rather than a hardware ceiling; optimising the processing stack (e.g., rewriting latency-critical modules in C++) would extend the usable frequency range and support additional concurrent users without changing the minimum frequency spacing. Four fixed 850 nm beacons at projection corners provide anchors for automatic homography calibration into Unity’s normalised canvas space $[0, 1]^2$.

Our Python pipeline detects infrared blobs, buffers each blob’s binary presence signal (2 s), estimates dominant modulation via FFT (± 3 Hz tolerance), maps frequency to flashlight ID, and applies Kalman filtering for robust tracking under brief occlusions. Tuples (*user_id*, *x*, *y*, *confidence*) stream to Unity via WebSocket at 70 fps; Unity resolves illuminated ROI(s) and routes corresponding spatial audio. End-to-end latency averages 18 ms (25 ms peak), below the 50 ms threshold for tightly coupled tangible feedback (Figure 2c).

2.6 Multi-User Coordination: Turning Conflicts into Social Moments

Overlapping illumination can confuse frequency estimation and raises questions about concurrent audio. We treat these moments as social opportunities rather than errors.

When two flashlights approach within ≈ 30 pixels in camera space (≈ 8 cm on the surface), FFT estimates may become ambiguous

Table 1: System bill of materials and cost estimate.

Component	Unit cost (USD)	Subtotal (USD)
IR flashlight ×6 (ESP32, 850 nm LED, 3D-printed housing, battery)	15	90
Camera + optical filter (400 fps global shutter + 850 nm narrowband)	65	65
IR anchor beacons ×4	2.50	10
Total (approx.)		~165
Commercial alternative (OptiTrack entry-level)		>5,000

due to signal superposition. We address this with *ID persistence*: each track caches its last confirmed frequency for 1.5 s and, if analysis degrades within this window, continues reporting the last-known identity while reducing confidence. This preserves continuity even when spotlights nearly overlap.

If two visitors illuminate the same ROI, Unity reduces both narration volumes to 80%, signalling proximity without forcing separation. In the field, visitors often read this attenuation as shared attention and it frequently triggered spontaneous conversation.

3 Implementation

Living Scrolls combines low-cost custom hardware with open-source software. Each flashlight uses an ESP32 to PWM-modulate an 850 nm near-infrared LED (≈USD 15 per unit including 3D-printed housing and battery). Sensing employs a global-shutter industrial camera (400 fps; ≈USD 60) with an 850 nm narrowband filter (USD 5) and four infrared anchor beacons (USD 10 total). The pipeline runs in Python 3.8 (OpenCV) and Unity 2021, communicating via WebSocket-Sharp; audio is produced in Audition (binaural stereo), and animations are hand-drawn (Procreate), looped (After Effects), and deployed via Spine 2D runtime. Table 1 details the bill of materials. Supporting six concurrent users costs ≈USD 165 total (excluding projector)—orders of magnitude lower than commercial optical tracking (OptiTrack entry-level: USD 5,000–15,000; >USD 100,000 for large spaces).

4 Field Deployment and Insights

We deployed Living Scrolls for five days at the Great Bao'en Temple Ruins Museum, a Buddhist heritage-site museum in Nanjing, China, projecting *The Kangxi Emperor's Southern Inspection Tour, Volume 11* [8]. Following an in-the-wild methodology [14], we recruited 20 expert reviewers (museum practitioners, heritage scholars, government representatives) and 73 general visitors (aged 20–40, friend-and-family groups). Data were collected through structured field-notes, informal exit interviews, and on-site video recordings, and synthesised thematically across both cohorts.

Emergent social dynamics. Visitors consistently formed *guide-learner* dyads—one person verbally directing another's flashlight toward points of interest—most pronouncedly in family groups, where parents scaffolded children's attention toward historically significant figures [2], a behaviour directly afforded by the shared projection. Several participants photographed and shared the experience on social media, suggesting propagation into networked publics.

Mean dwell time substantially exceeded typical static-painting viewing (~30–60 s [4]): several visitors engaged for >8 minutes, iteratively switching ROIs to assemble what one participant described as a narrative “puzzle”. We interpret these patterns as evidence that the visible-spotlight design sustains collective attention even when audio is individually routed—resolving, at least partially, the personalisation–presence tension identified in §2.1.

Usability friction. Two issues surfaced consistently. First, visitors expected audio to switch immediately upon entering a new ROI, conflicting with our 1.5 s dwell lock—a tension between *narrative coherence* (preventing fragmented listening) and *exploratory immediacy* (a responsive-flashlight mental model). Second, infrared reflections from polished floors produced occasional “ghost” spots outside the painting boundary, requiring on-site threshold adjustment. Both issues reveal that deployment context requires site-specific calibration that a one-click template cannot fully automate.

Expert assessment. Professional reviewers endorsed the system's low-cost reproducibility as enabling adoption beyond well-funded institutions, and noted the cross-cultural accessibility of the flashlight metaphor, with both Chinese and European visitors showing near-zero learning curve.

5 Discussion

Living Scrolls addresses a structural gap in the *Revealing Flashlight* paradigm [13]: prior systems lacked user–spot association, making personalised audio impossible in shared space. Our frequency-encoded approach resolves this at low cost (≈USD 165), but deployment surfaced tensions that generalise beyond this system.

Personalisation and presence as complementary, not competing. A recurring assumption in museum technology is that personalisation necessarily fragments collective experience—audio guides being the canonical example [9]. Our findings challenge this framing. The visible-spotlight design ensured that individual exploration remained socially legible: visitors could see where others were looking, which directly catalysed guide–learner dynamics and spontaneous conversation. Headphones privatised listening without privatising attention, supporting social behaviours similar to conventional audio guides where private listening coexists with engagement [3, 5].

Embodied metaphor as cross-cultural scaffold. The near-zero learning curve observed across both Chinese and European visitors—without instruction—points to the flashlight's power as a culturally portable embodied metaphor. Unlike touch interfaces or gesture-based systems that require surface instrumentation or calibration to user morphology, the flashlight externalises intent

through physical direction, making interaction legible to both the actor and bystanders. This has implications for heritage installations targeting diverse international audiences.

Limitations and future directions. Three challenges remain open. First, the 1.5 s ROI lock, while reducing accidental triggers, conflicts with visitors’ “follow-the-flashlight” mental model; progressive audio crossfades may better reconcile narrative coherence with exploratory immediacy. Second, infrared reflections from polished or glass surfaces require site-specific calibration that resists full automation—future work should develop environmental scanning tools to flag reflection risks during setup. Third, our analysis draws on fieldnotes and informal interviews; a structured comparative study would better isolate the contribution of frequency-encoded personalisation from the novelty effect of the installation itself. We will release an open-source deployment package including hardware schematics, software pipeline, and calibration utilities to support replication across resource-constrained institutions [1, 12].

6 Conclusion

Living Scrolls combines frequency-encoded infrared tracking with multi-user flashlight interaction to deliver personalised spatial audio on a shared projected painting, providing museums with a low-cost ($\approx$ USD 165), cross-cultural, reproducible immersive heritage format. A five-day in-the-wild deployment demonstrated technical feasibility and social value: experts recognised its novelty and scalability, while visitors’ spontaneous collaboration evidenced strong ecological validity. By open-sourcing the system, we lower the economic barrier to immersive cultural heritage, enabling local museums and community sites to make archived histories experientially present; future work will release a deployment toolchain and training materials to support wider adoption.

References

- [1] Antonino Cosentino. 2015. Practical notes on ultraviolet technical photography for art examination. *Conservar Patrimônio* 21 (2015), 53–62.
- [2] Kevin Crowley and Maureen Callanan. 1998. Describing and supporting collaborative scientific thinking in parent-child interactions. *Journal of Museum Education* 23, 1 (1998), 12–17.
- [3] Brian Elliston and Elizabeth FitzGerald. 2012. Encouraging museum visitor engagement using spontaneous talk-in-interaction audio guides. (2012).
- [4] John H Falk and Lynn D Dierking. 2016. *The museum experience revisited*. Routledge.
- [5] Rebecca E Grinter, Paul M Aoki, Margaret H Szymanski, James D Thornton, Allison Woodruff, and Amy Hurst. 2002. Revisiting the visit: Understanding how technology can shape the museum visit. In *Proceedings of the 2002 ACM conference on Computer supported cooperative work*. 146–155.
- [6] Jan Gugenheimer, Christian Mai, Mark McGill, Julie Williamson, Frank Steinicke, and Ken Perlin. 2019. Challenges using head-mounted displays in shared and social spaces. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–8.
- [7] Eva Hornecker and Luigina Ciolfi. 2022. *Human-computer interactions in museums*. Springer Nature.
- [8] Wang Hui et al. 1691–1698. Kangxi Emperor’s Southern Inspection Tour, Scroll Eleven. <https://www.dpm.org.cn/collection/paint/228301.html>. Ink and colour on silk, 67.8 × 2612 cm. The Palace Museum, Beijing. Digital image used under fair scholarly use for academic illustration..
- [9] Rachel Hutchinson and Alison F Eardley. 2021. Inclusive museum audio guides: ‘guided looking’ through audio description enhances memorability of artworks for sighted audiences. *Museum Management and Curatorship* 36, 4 (2021), 427–446.
- [10] K Jaebker and G Bowman. 2015. Context is king: Using indoor-location technology for new visitor experiences. *MW2015: Museums and the web 2015* (2015).
- [11] Gaea Leinhardt, Kevin Crowley, and Karen Knutson. 2003. *Learning conversations in museums*. Taylor & Francis.
- [12] David Myers, Alison Dalgity, and Ioannis Avramides. 2016. The Arches heritage inventory and management system: a platform for the heritage field. *Journal of cultural heritage management and sustainable development* 6, 2 (2016), 213–224.
- [13] Brett Ridel, Patrick Reuter, Jeremy Laviolle, Nicolas Mellado, Nadine Couture, and Xavier Granier. 2014. The revealing flashlight: Interactive spatial augmented reality for detail exploration of cultural heritage artifacts. *Journal on Computing and Cultural Heritage (JOCCH)* 7, 2 (2014), 1–18.
- [14] Yvonne Rogers and Paul Marshall. 2017. *Research in the Wild*. Morgan & Claypool Publishers.
- [15] Susie Wilkening. 2025. *Social Experiences and Museums, Part One: A 2025 Annual Survey of Museum-Goers Data Story*. American Alliance of Museums. <https://www.aam-us.org/2025/10/24/social-experiences-and-museums-part-one-a-2025-annual-survey-of-museum-goers-data-story/> Alliance Blog (Data Story).
- [16] Lisa Yanagihara, Hiroaki Ishida, Shigenori Inagaki, and Fusako Kusunoki. 2025. Muse-Torch: Interactive Museum Guide System Using a Pico-Projector for Children. In *Proceedings of the 24th International Conference on Mobile and Ubiquitous Multimedia*. 463–466.
- [17] Neng-Hao Yu, Li-Wei Chan, Seng Yong Lau, Sung-Sheng Tsai, I-Chun Hsiao, Dian-Je Tsai, Fang-I Hsiao, Lung-Pan Cheng, Mike Chen, Polly Huang, et al. 2011. TUIC: enabling tangible interaction on capacitive multi-touch displays. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 2995–3004.